Information Retrieval Notes

Q1. Explain Basic concepts of IR.

Information retrieval can be defined as software program that deal with organization, storage, retrieval and evaluation of information from document reporitories particularly textual information.

Important terminologies in IR :

a) Data :
   - Raw representation of collections from various sources or single source is data.
   - Data may or may not contain information.
   - Data can be either quantitative or qualitative.

b) Information :
   - Data which has significance indecision making in some action if information.
   - Information is derived from data.

c) Knowledge :
   - When information is processed, linked & stored by human or by m/c then information become knowledge.
   - As more & more information is given, knowledge can increase.

d) Retrieval :
   - From already present data trying to search or find some particular pattern given by user is retrieval.
   - Retrieval can be based on search words, keys or any other specific data.

Q2. Data vs information retrieval.

	Parameters	Data retrieval	Information retrieval
1	Matching	Exact match	Partial match
2	Model	Deterministic	Probabilistic
3	Classification	Monothetic	Polythetic
4	Query language	Artificial	Natural
5	Query specification	Complete	Incomplete

Q3. Explain Trends in IR.

a) Faceted search :
   - Faceted search is technique that allows for integrated search and navigation experience by allowing users to explore by filtering available information.
   - This search is used often in ecommerce websites and applications.

b) Social search :
   - Social search is new phenomenon facilitated by recent web technologies.
   - Collaborative social search involves different ways for active involvement in search related use of networks involving social data mining.

c) Conversational search :
   - Conversational search is an interactive and collaborative information finding interaction.
   - If uses semantic retrieval model with natural language understanding to provide users with faster and relevant search.

Issues in IR :

a) Document and query indexing :
   Main goal of document and query indexing is to find important meaning and creating an external representation.
b) Query evaluation :
   Uncertainty : The available representation does not typically reflect true semantic of objects such as images, videos, etc.
   Vagueness : The information that user requires lacks clarity, it is only vaguely expressed in query, feedback or user action.
c) System evaluation : System evaluation tells about importance of determining the impact of information given on user achievement.

Q4. Explain Document representative.

- An information retrieval system deals with representation storage, organization and access of information.
- The input for this system is list of documents which contain information and query given by user.
- Main problem here is to obtain representation of each document and query suitable for computer to use.
- At first step documents are converted into its representation. The documents in its natural language format are one of the representations.
- But if use store these document in natural language space requirement increases as well as time only representation of document.
- A document representative could be list of extracted words considered to be significant. Those are called as keywords.
- Information retrieval system are two types, one is manual retrieval system and automatic retrieval system.

Q5. Explain Information retrieval process

- User query formation : Process begins with user formulating query or set of keywords that describes information they want.
- Query processing : System processes user's query to understand search criteria.
- Indexing : Information in database is preprocessed and indexed.
- Searching : The system uses the processed query and index to search for relevant documents in database.
  Various techniques & algorithms are used for it.
- Ranking and relevance scoring : Retrieved documents are ranked based on their relevance to user's query.
- Retrieval & presentation : Top ranked documents are retrieved and presented to user.
- User feedback : User may provide feedback on relevance of presented results.
- Interactive process : IR is interactive process.
  Based on user feedback, system tries to enhance accuracy & relevance of results.

Q6. Issues in IR & refinement.

a) Indexing efficiency :
   - Issue : Generating & maintaining an index for large dataset can be time consuming.
   - Ref. : Use efficient indexing techniques such as inverted indexes, compression, distributed indexes.

b) Query processing speed :
   - Issue : Processing user queries in real-time, especially for large datasets, can be challenging.
   - Ref. : Important query optimization techniques, caching mechanisms & parallel processing.

c) Scalability :
   - Issue : Scaling system to handle growing volume of data and increasing user queries can lead to performance degradation.
   - Ref. : Employ distributed architectures, clod-based solutions, horizontal scaling to handle larger datasets.

d) Resource management :
   - Issue : Efficiently managing system resources, memory & processing power is crucial for performance.
   - Ref. : Important resource-aware algorithm, monitor system performance, allocate resources.

e) Relevance ranking :
   - Issue : Determining relevance of document to user query accurately is complex task.
   - Ref. : Continuously refine and update the ranking algorithms based on user feedback & user needs.

Q7. Information retrieval in digital libraries.

- Information retrieval in digital libraries involves the organization, storage, retrieval of digital information within the library setting.
- Digital libraries leverages technology to provide user's access to vast digital sources such as text, images, audio, video, multimedia content.

Key aspects :

a) Content digitization :
   - Digital libraries involve conversion of physical materials into digital formats.
   - This process involves digitizing books, journals and other resources.

b) Metadata creation & indexing :
   - Metadata or data about data is crucial for organizing and retrieving digital content.
   - Each digital resource is associated with metadata.

c) Search interfaces :
   - Digital libraries provide search interfaces that allow users to input queries and retrieve relevant information.
   - Search interfaces can range from basic keyword boxes to advanced options such as filters.

d) Information retrieval models :
   - Various information retrieval models are employed to rank and retrieve relevant documents based on user queries.

- Models such as vector space model, Boolean model, and probabilistic model may be used to enhance performance.

e) Multimedia retrieval :
   - Digital libraries often host diverse range of multimedia content, including images, audio, video.
   - Retrieving multimedia content involves techniques such as content based image retrieval, speech recognition, video analysis.

Q8. Difference between Boolean Model and Vector Model.

	Boolean model	Vector model
1	Here only presence or absence of term is modeled.	Here weightage of term is modeled.
2	Efficient in calculation.	Requires more calculations than Boolean model.
3	There is no way to model global dependency.	Global dependency of document is represented.
4	Dependencies between documents are partially represented.	Dependencies between documents are better represented.
5	Less popular than vector model.	Useful in many apps.

Q9. Difference between Natural Network Retrieval and Fuzzy Set Retrieval.

	Natural network retrieval	Fuzzy set retrieval process
1	Interactive activation model of word perception.	Fuzzy document representation based on Boolean model.
2	Query is represented from tags.	Query is considered as weighted words or terms.
3	Neural n/w is complex.	It is simpler than.
4	It helps to perform prediction.	It helps to perform pattern recognition.
5	Difficult to extract knowledge.	Knowledge can be easily extracted.

Q10. Taxonomy of IR models & IR modelling technique.

Information retrieval models :

- Classical models :
  a) Boolean model
  b) Vector space model
  c) Cluster based model
- Probabilistic models :
  a) Binary independence model
  b) Language model
- Neural models :
  a) Recurrent neural networks
  b) Transformer-based models
- Relevance feedback models :
  a) Rocchio algo
  b) Markov model.

Q11. Explain Vector space model.

- Vector space model is classical and widely used IR model that represent documents & queries as vectors in high-dimensional space.
- Key concepts :
  i. Term document matrix : Term document matrix captures frequency or weight of each term in each document.
  ii. Vector representation : Each document & query is represented as vector in term space.
  iii. Cosine similarity : Retrieval is performed by calculating cosine similarity between the query vector & the document vectors.
- Process :
  i. Document indexing : Create a term-document matrix, where row corresponds to term & column corresponds to document and matrix element represent weight.
  ii. Query vectorization : Represent user query as vector where each dimension corresponds to term and vector's components represent query's term weights.
  iii. Cosine similarity calculation : Calculation the cosine similarity between query vector & each document vector.
  iv. Ranking & retrieval : Rank documents & then retrieve top-ranked documents as search results.
- Advantage :
  i. Allows for flexible weighting schemes, such as TF-IDF.
  ii. Effective for capturing the semantic relationships between terms & documents.
- Limitations :
  i. Assumes term dependence, which may not hold in all cases.
  ii. Does not explicitly model word semantics or context.

Q12. Explain Boolean model.

- Boolean is one of the earliest and simplest models used in information retrieval.
- It is based on Boolean algebra and allows users to formulate queries using logical operators.
- Primarily used for text-based IR system.

Key components :

i. Boolean operators : Boolean model uses three logical operators to combine terms in query.
   a) AND (conjunction) : Retrieve documents containing all specified terms. Represented by operator "AND".
   b) OR(disjunction) : Retrieves documents containing at least one of the specified terms represented by operator "OR".
   c) Not (exclusion) : Excludes documents containing specified term. Represented by operator "NOT"

ii. Query formation :
   a) Users constructs queries using combination of terms and Boolean operators.

b) Ex. Query 1 : "Information and retrieval".
   Query 2 : "Database or system".
   Query 3 : "Boolean not model".

iii. Binary representation :
   - Each document and query is represented as binary vector.
   - The value of each document is 1 if term is present in document or query or else 0.

iv. Document retrieval :
   - Boolean model retrieves documents that satisfy logical conditions specified in query.
   - For "AND" retrieves documents containing all terms.
     "OR" retrieves documents containing at least one term.
     "NOT" excludes documents containing specified terms.

Advantages :

a) Simplicity : i) Straight forward ii) Easy to understand.
b) Precision : results are precise.

Limitations :

a) No ranking : doesn't rank documents by relevance .
b) Vocabulary mismatch.

Q13. Explain the significance of user relevance feedback in an IR system.

- User relevance feedback is crucial aspect of IR system and its significance lies in improving effectiveness and user satisfaction of retrieval process.
- Here are some points.

1. Improved precision and recall :
   a) User relevance feedback allows users to provide explicit feedback on relevance of retrieved documents.
   b) This feedback can be used to refine the search algorithm improving precision & recall.
2. Personalization :
   a) Relevance feedback enables the system to learn from user's preferences over time.
   b) By considering user's feedback on previous search results IR system can adapt its ranking with individual user's information needs.
3. Contextual understanding :
   a) Users can provide additional context or information about their search intent through relevance feedback.
   b) This helps system better understand user's requirements.
4. Reduced information overload :
   a) In large information spaces, user often face information overload, making it challenging to find relevant content.
   b) Relevance feedback helps in filtering out irrelevant information & presenting users with more focused results.

Q14. Explain Structured text retrieval.

a) Structured text retrieval involves retrieving relevant information from structured documents or databases based on user queries.
   Process :

- Document structuring :
  Define document structure : identify structure of documents or database entries.
- Data modeling : Represent the structured data using an appropriate data model.

b) Indexing :
- Identify key fields : Determine which fields or attributes are important for retrieval.
- Create index : Generate an index that maps terms in these key fields to corresponding documents or record identifiers.

c) Query parsing :
- User query analysis : Parse & analyze user queries to identify key terms.
- Map to structured query : Convert user queries into structured queries that match underlying data model.

d) Query processing :
- Optimization : Optimize structured query for efficient retrieval.
- Execute query : Use optimized query to retrieve relevant records from structured data.

e) Ranking :
- Relevance ranking : IF there are multiple matching records rank them on relevance.

f) Result presentation :
- Format results : Present the retrieved information in a format that is understandable & relevant to user.

g) User feedback :
- Relevance feedback : allow users to provide feedback on the relevance of retrieved results.

h) Continuous improvement :
- Monitoring & evaluation : Continuously monitor performance of structured text retrieval system. Evaluate user satisfaction.
- Iterative refinement : Use the insights gained from monitoring and evaluation to iteratively refine the system.

Q15. Explain types of queries & pattern making queries.

a) Boolean queries :
- Use logical operators (AND, OR, NOT) to combine or exclude terms.
- Ex. "information AND retrieval".

b) Phrase queries :
- Search for documents containing a specific phrase or sequence of words.
- Ex. Artificial intelligence.

c) Range queries :
- Search for documents that contain terms within specific numerical of alphabetical range.
- Ex. "Prince : (10 to 50) could match documents with prices between 10 and 50.

d) Structured queries :
- Use specific syntax or structure to represent complex relationships between terms.
- Ex. "title : data AND author : smith".

Pattern matching queries :

i. Pattern matching queries in IR involve search for documents or data that match specific pattern or template.
   Use cases :
   - Structured data extraction
   - Information validation
   - Template-based search
ii. Elements of pattern matching queries :
   a) Literal character :
      - Specific characters that must match exactly.
      - Ex. In pattern " (email protected)" characters "email protected" are literals.
   b) Wildcards :
      - Place holder characters that match any character.
      - Common wildcards '* ' and '?'
   c) Character classes :
      - Define set of character that can match at specific position.
      - For instance "(0-9)" can represent any single digit.
   d) Anchors :
      - Specify position of pattern within document.
      - Ex. "^(email protected)" might match "email protected" at the beginning of line.
   e) Example :
      - Data pattern :
        Pattern : "DD-MM-YYYY"
        Query : extract dates in format DD-MM-YYYY from documents.
        Result : retrieves instances of dates in specified formats.
      - Phone no. pattern :
        Pattern : "(###) ###-####"
        Query : extract phone no. in format "(###) ###-####"
        Result : retrieves instances of phone no matching specified patterns.
   f) Challenges :
      - Variability
      - False positives

Q16. Explain Keyword based querying.

- A query is expression of user's information requirement.
- In its most basic form, query is mode up of keywords and documents that contain keywords are searched for.
- Keyword queries are popular because they are simple to express & allow for quick ranking.
- These keywords are connected by logical AND operator.

  Types :
  Simple word queries
  Context queries
  Boolean queries

- Keyword based querying in IR is common method for users to search and retrieve information from collection of documents or database.

- Keyword based querying is widely used in web search engines, databases & retrieval systems.

  Types :

  a) Boolean queries :
     i. Def^n : Boolean queries use logical operators (AND, OR, NOT) to continue keywords & specifying relationship between them.
     ii. Ex. : “information AND retrieval”, “database OR management”.
  b) Phrase queries :
     i. Def^n : Phase queries involve searching for specific sequence of words, treating them as single unit.
     ii. Ex. : “data science” machine learning algorithm.
  c) Range queries :
     Def^n : Range queries involve searching for terms which a specific range.
     Ex. : “price : (50 to 100)”, “ data : (2022-01-01 to 2022-12-31)”.

Structural queries :

- Structural queries in information retrieval refers to search queries that takes into account structural elements or features of documents in addition to their content.
- These queries are useful to retrieve document based on how they are organized, formatted or linked within document collection.
- Structural queries can be particularly useful when the document structural itself is important, such as document databases, XM2 collections or web documents.
- In structural queries the focus is not just on textual content of documents but also on their structural elements which can include heading sections, tables, links, metadata, etc

Types :

a) Fixed queries
b) Hierarchical queries.
c) Link queries
d) Tag-based queries

Q17. Racchio method for query expansion.

- The racchio method is relevance feedback technique use in IR systems, such as search engines to improve precision of search results.
- It expands query based on feedback from user about relevance of initial search results.
- Method :
  i. Initial query : User enters initial query to search for info
  ii. Initial retrieval : Search engine retrieves set of documents to user’s query.
  iii. User feedback : User interacts with results & provides feedback.
  iv. Document vectors : Racchio method represents each document & query as vectors in high dimensional space.
  v. Relevance feedback : Racchio algo calculates new query based on feedback received.
  vi. Query expansion : The adjusted query vector is used to create new expanded query. Repeated search : Search engine performs new search with expanded query.
  vii. Improved results :

- Key parameters :
    i. Query vector
    ii. Relevance feedback
    iii. Feedback iterations.

Q18. Explain Query expansion through local content analysis :

- Query expansion through local content analysis is technique used to improve effectiveness of search queries in information retrieval system.
- It involves analyzing content of documents retrieved by additional terms.
- Working of query expansion through LCA :
    i. Initial query
    ii. Initial retrieval
    iii. Local content analysis
    iv. Term extraction
    v. Query expansion
    vi. Revised search
    vii. Improved results
- Key points :
    i. Relevance threshold
    ii. Scalability
    iii. Feedback loop
    iv. User control

Q19. Explain Inverted files.

- Inverted files or inverted index is fundamental data structure used in information retrieval systems.
- In search engines to optimize and speed up the process of searching for documents inverted files used.
- It is called as "inverted" because it reverses typical relationship between documents & words.
- Instead of listing which words are in each document, it lists which document contain each word.

Components :

i. Dictionary : Dictionary is list of unique terms found in collection of documents.
ii. Posting list : posting list contains document identifiers where each term appears.

Advantages :

i. Efficient retrieval
ii. Reduced storage
iii. Scalability

Disadvantages :

i. Storage overhead
ii. Indexing overhead

Use cases :

i. Search engines
ii. Information retrieval systems
iii. Text analysis

Example :

- Forward index
  i. Document 1 : “Information retrieval system”
  ii. Document 2 : “Data mining techniques”
- Inverted index
  i. “information” = (Document 1)
  ii. “retrieval” = (Document 1)
  iii. System = (Document 1)
  iv. Data = (Document 2)
  v. Mining = (Document 2)
  vi. Techniques = (Document 2)
- (process) Steps :
  i. Tokenization
  ii. Normalization
  iii. Stop word removal
  iv. Stemming
  v. Building inverted index

Q20. Explain Signature file structure.

- A signature file structure is a data structure used in searches.
- Signature files provide compact representation of documents.

- Key components
  i. Documents or records :
     Collections of documents or records that need to be searched.
  ii. Signatures :
     Each document is associated with signature which is compact representation of documents content.
  iii. Bit vectors :
     Signatures are typically implemented as bit vectors, where each bit represents presence or absence of specific term.
- How signature file work
  i. Document representation
  ii. Bit vector
  iii. Query signature
  iv. Matching
  v. Result set
- Ex.
  Consider collection of books & query looking for “science” and “fiction”. Each book is associated with signature.
  Book 1 (science & fiction) : 1100
  Book 2 (fiction) : 0100
  Book 3 (science) : 1000
  Book 4 (non relevant) : 0001

Q21. Explain Sequencial searching.

- Sequencial searching is simple method for finding a particular item in collection of data.
- Also called as linear searching.

- It required info is found or entire document has searched.
  Process :
  i. Start at beginning
  ii. Compare with target
  iii. Stop on match
  iv. Continue or stop

- Time complexity : O(n)

- Limitations :
  i. Inefficiency for large collections
  ii. Not suitable for real time requirements
  iii. Not ideal for sorted data

- Applications :
  i. Textual data search
  ii. Small databases

- Example :
  Scenario : Consider document collection with titles. You want to find document with title " IR basics".
  Sequencial search :
  Start from first document, compare title with IR basics until match is found or end of collection is reached.

Q22. Explain TF-IDF weighting.

- TF-IDF stands for term frequency-inverse document frequency.
- It is a numerical statistic used in information retrieval, NPL & text mining to evaluate importance of term in document.
- TF-IDF weighting helps determine which terms are significant within individual document & across corpus.
- TF-IDF weighting is powerful tool for analyzing & ranking documents based on their content,
- Term frequency :
  TF (t_1d) = (No. of times term t appears in document d)
  (total No. of terms in document d)
- Reserves document frequency :
  IOE (t) = log-e (total no. of document s containing term t)
- TF-IDF score :
  TF-IDF(t_1d) = TF (t_1d) * IDF(t)
- Key points :
  i. Importance of rare terms
  ii. Scaling
  iii. Document retrieval
  iv. Information retrieval
  v. Text classification
  vi. Preprocessing

a) Term frequency (TF)
- Term frequency measure how often a term appears in document.

- It is calculated as ratio of no. of occurrences to total no. of terms in document.
- Formula : $TF(t_1 d) = \frac{\text{no. of times term t appears in document d}}{\text{total no. of terms in document d}}$

b) Inverse document frequency (IDF)

- IDF measuring rarity or uniqueness of term across collection of document.
- It is calculated as logarithm of ratio of total no of document to no. of documents containing term.
- Formula : $IDF(t_1 d) = \left(\frac{\text{total no. of document in D + 1}}{\text{no. of document containing t}} \right) \log$

c) If-IDF weighting :

- TF-IDF weight for term t in document d by multiplying TF& IDF.
- TF-IFD $(t_1 d_1 n)$ = TF $(t_1 d)$ * IDF $(t_1 D)$

d) Applications :

1. Information retrieval
2. Document clustering
3. Text summarization.

Q23. Explain Evaluation process – IR.

- Evaluation process of IR system typically involves system-based evaluation and user-based evaluation.
- Each type of evaluation has its purpose & provides valuable insights.

System –based evaluation :

- System-based evaluation include assessing precision, recall, ranking algorithms or other technical aspects of IR.
- Choose matric that reflect technical performance common matric includes precision, recall, F1 score, mean average precision (MAP), etc.
- Human assessors evaluate the relevance of documents to specific queries based on predefined criteria.
- Document the system-based evaluation process, methodologies, data sets, technical metrics & results.

User based evaluation :

- Define goals related to user experience, satisfaction & ease of use.
- Choose matrics that capture user satisfaction, such as user surveys, feedback ratings or usability measures.
- Analyze user interactions, such as click rates, d well time, other behaviors.
- Gather feedback from users through surveys, interviews or observations to understand their satisfaction & experience.
- Document the use-based evaluation process, methodologies, user feedback & satisfaction metrics.

Q24. Explain Multimedia information retrieval.

- Multimedia IR is field of study that focuses on retrieving relevant info from multimedia databases.

- Unlink traditional IF that deals with text based docs, multimedia IR extends scope to various media types, including images, audio, video, other multimedia content.

1. Media types :
   Image : retrieval of images based on visual content, features, metadata.
   Audio : retrieval of audio based on pitch, tempo.
   Video : retrieval of video based on visual & auditory components.

2. Key techniques & approaches :
   Content based retrieval
   Text based retrieval
   Multimodal fusion
   ML

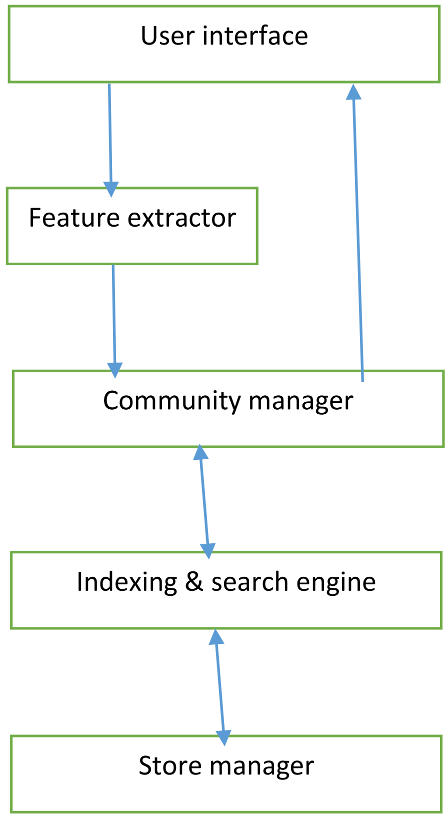
3. Evaluation metrics :
   Precision & recall : measures accuracy
   Mean avg precision (MAP) : avg precision values
   Fl score : balancing precision & recall.

4. Applications :
   Content based image retrieval (CBIR)
   Speech & audio retrieval
   Video retrieval
   Multimodal search
5. Challenges :
   Content diversity (diverse content types)
   Semantic GAP (address GAP between features)
   Scalability (large scale multimedia dbs)

MIRS architecture :

- MIRS consist functional modules – added to extend, deleted or replaced.
- Characteristics : a) consisting of servers & clients.

c) Results from large size of multimedia data.



1. User interface :
   Insertion of new multimedia items & retrieval.
2. Feature extractor :
   These items can be stored files or input from devices.
3. Communication manager :
   The contents or features of multimedia items one extracted either automatically or semi-automatically.
4. Indexing & search engine :
   At server, feathers are organized according to a certain indexing scheme for retrieval.
5. Storage manager :
   The indexing information and original items are stored appropriately.

Q25. Explain Distributed IR.

- Distributed IR is field of study the focuses on development of information retrieval systems distributed across multiple nodes or servers.
- Unlike traditional IR systems, where a single server handles all queries & documents, DIR leverages distributed computing to enhance scalability, fault tolerance & efficiency.

A. Key concepts :
   - Document portioning :
     Distributing documents across different nodes based on predefined criteria.
   - Query partitioning :
     Dividing user queries into sub queries that can be processed independently by different nodes.

- Index partitioning :
  Splitting inverted index into partitions, each managed by different mode.

B. System architecture :
  - Node architecture : each node is distributed including components for query processing, index management, etc.
  - Communication layer : efficient communication channels & protocol are established.
  - Load balancing : mechanisms for distributing queries & workload.

  Challenges :
  1. Network latency
  2. Data skew
  3. Consistency overhead

  Applications :

  1. Web search engines
  2. Enterprise search
  3. Cloud based systems

  Advantages :

  1. Scalability : handles large volume of data by distributing workload
  2. Fault tolerance : handle failure
  3. Efficiency : parallel processing & distribution improves response time.

Q26. Difference between Suffix Trees and Suffix Arrays.

	Suffix trees	Suffix arrays
1	Suffix trees are dynamic data structures & can be efficiently updated when the underlying string is modified.	Suffix arrays are static data structures & less suitable for handling dynamic text updates.
2	It consumes more memory than suffix arrays.	It consumes less memory.
3	It is less space efficient.	It is more space efficient.
4	Building a suffix tree can be more time consuming.	It is simpler array based data structure that stores the starting position of all the suffixes of string in sorted order.

Q27. Explain Latent semantic indexing model.

- Latent semantic indexing is also called as latent semantic analysis (LSA).
- It is a natural language processing & IR technique that aims to discover latent semantic structure of collection of documents.

Steps :

- Document term matrix : create document term matrix (DTM)
- Term weighting : apply term weighting technique (ex. TF-INF)
- Single value decomposition : perform SVD on DTM.

- ➢ Dimensionality reduction : reduce dimensionality of data.
- ➢ Conceptual representation : each document & term is represented by vector
- ➢ Query processing : process user query.
- ➢ Ranking & retrieval :

Advantages :

1. Semantic understanding :
   LSI goes beyond exact matching of terms & captures underlying semantic relationship between words.
2. Dimensionality reduction :
   LSI reduces dimensionality of term-document space, allowing for more storage & processing of large datasets.
3. Handling synonymy & polysemy :
   LSI addresses challenges of synonymy (multiple words with similar meanings) & polysemy (single words with multiple meaning).
4. Improving IR :
   LSI enhances information retrieval by providing more accurate & relevant results.

5. Reducing noise :
   LSI helps to reduce impact of noise in textual data, such as irrelevant terms or variation in language.